

WMO FM 301 NetCDF Validation Report of Sample_file.nc

Section Summaries

Section	Pass	Fail Mandatory	Fail optional
Global_Attributes	7	5	0
Global_Ancillary_variables	11	9	0
sweep_variables	15	16	0
data_variables	4	3	0
radar_parameters	0	0	0
radar_calibration	0	0	0
Overall	37	33	0

Detailed Results

Group	Name	Available	Expected Dtype	Actual Dtype	Expected Value	Actual Value	Requirement	Result
Global_Attributes	instrument_name	Yes	string	str	None	None	Mandatory	pass
Global_Attributes	institution	Yes	string	str	None	None	Mandatory	pass
Global_Attributes	references	Yes	string	str	None	None	Mandatory	pass
Global_Attributes	source	Yes	string	str	None	None	Mandatory	pass
Global_Attributes	history	Yes	string	str	None	None	Mandatory	pass
Global_Attributes	comment	Yes	string	str	None	im/exported using xradar	Mandatory	pass
Global_Attributes	platform_is_mobile	No	string	None	None	None	Mandatory	fail_mandatory
Global_Attributes	Conventions	Yes	string	str	CF-1.8, WMO CF-1.0	None	Mandatory	fail_mandatory
Global_Attributes	wmo__cf_profile	No	string	None	FM 301-2022	None	Mandatory	fail_mandatory
Global_Attributes	title	Yes	string	str	None	None	Mandatory	pass
Global_Attributes	wmo__data_category	No	string	None	None	None	Mandatory	fail_mandatory
Global_Attributes	wmo__data_policy	No	string	None	['core', 'recommended']	None	Mandatory	fail_mandatory
Global_Ancillary_variables	volume_number	Yes	int	int64	None	0	Mandatory	fail_mandatory
Global_Ancillary_variables	time_coverage_start	Yes	string		None	2013-05-10T00:00:06Z	Mandatory	pass

Group	Name	Available	Expected Dtype	Actual Dtype	Expected Value	Actual Value	Requirement	Result
Global_Ancillary_variables:time_coverage_start	units	No	string	None	seconds since .	None	Mandatory	fail_mandatory
Global_Ancillary_variables:time_coverage_start	calendar	No	string	None	['gregorian', 'standard']	None	Mandatory	fail_mandatory
Global_Ancillary_variables:time_coverage_start	standard_name	No	string	None	time	None	Mandatory	fail_mandatory
Global_Ancillary_variables	time_coverage_end	Yes	string		None	2013-05-10T00:03:14Z	Mandatory	pass
Global_Ancillary_variables:time_coverage_end	units	No	string	None	seconds since .	None	Mandatory	fail_mandatory
Global_Ancillary_variables:time_coverage_end	calendar	No	string	None	['gregorian', 'standard']	None	Mandatory	fail_mandatory
Global_Ancillary_variables:time_coverage_end	standard_name	No	string	None	time	None	Mandatory	fail_mandatory
Global_Ancillary_variables	latitude	Yes	double	float64	None	50.856633	Mandatory	pass
Global_Ancillary_variables:latitude	units	Yes	string	str	degrees_north	degrees_north	Mandatory	pass
Global_Ancillary_variables:latitude	standard_name	Yes	string	str	latitude	latitude	Mandatory	pass
Global_Ancillary_variables	longitude	Yes	double	float64	None	6.379967	Mandatory	pass
Global_Ancillary_variables:longitude	units	Yes	string	str	degrees_east	degrees_east	Mandatory	pass
Global_Ancillary_variables:longitude	standard_name	Yes	string	str	longitude	longitude	Mandatory	pass
Global_Ancillary_variables	altitude	Yes	double	float64	None	116.7	Mandatory	pass
Global_Ancillary_variables:altitude	units	Yes	string	str	metres	meters	Mandatory	fail_mandatory
Global_Ancillary_variables:altitude	standard_name	Yes	string	str	height_above_reference_ellipsoid	altitude	Mandatory	fail_mandatory

Group	Name	Available	Expected Dtype	Actual Dtype	Expected Value	Actual Value	Requirement	Result
Global_Ancillary_variables	platform_type	Yes	string		['fixed', 'vehicle', 'ship', 'aircraft', 'aircraft_fore', 'aircraft_aft', 'aircraft_tail', 'aircraft_belly', 'aircraft_roof', 'aircraft_nose', 'satellite_orbit', 'satellite_geostat']	fixed	Mandatory	pass
Global_Ancillary_variables	instrument_type	Yes	string		['radar', 'lidar']	radar	Mandatory	pass
sweep_variables	sweep_0/time	Yes	double	float64	None	0.0151515	Mandatory	pass
sweep_variables:sweep_0/time	units	Yes	string	str	seconds since .	seconds since 2013-05-10T00:00:06+00:00	Mandatory	pass
sweep_variables:sweep_0/time	calendar	Yes	string	str	['gregorian', 'standard']	proleptic_gregorian	Mandatory	fail_mandatory
sweep_variables:sweep_0/time	standard_name	Yes	string	str	time	time	Mandatory	pass
sweep_variables	sweep_0/range	Yes	float	float32	None	125.0	Mandatory	pass
sweep_variables:sweep_0/range	units	Yes	string	str	metres	meters	Mandatory	fail_mandatory
sweep_variables:sweep_0/range	standard_name	Yes	string	str	projection_range_coordinate	projection_range_coordinate	Mandatory	pass
sweep_variables:sweep_0/range	long_name	Yes	string	str	range_to_measurement_volume	range_to_measurement_volume	Mandatory	pass
sweep_variables:sweep_0/range	axis	Yes	string	str	cradial_range_coordinate	radial_range_coordinate	Mandatory	pass
sweep_variables:sweep_0/range	spacing_is_constant	Yes	string	str	None	true	Mandatory	pass
sweep_variables:sweep_0/range	meters_to_center_of_first_gate	Yes	float	float32	None	125.0	Mandatory	pass
sweep_variables:sweep_0/range	meters_between_gates	Yes	float	float32	None	250.0	Mandatory	pass
sweep_variables	sweep_0/frequency	No	float	None	None	None	Mandatory	fail_mandatory
sweep_variables:sweep_0/frequency	units	No	string	None	s-1	None	Mandatory	fail_mandatory
sweep_variables:sweep_0/frequency	standard_name	No	string	None	radiation_frequency	None	Mandatory	fail_mandatory
sweep_variables	sweep_0/sweep_number	Yes	int	int64	None	0	Mandatory	fail_mandatory

Group	Name	Available	Expected Dtype	Actual Dtype	Expected Value	Actual Value	Requirement	Result
sweep_variables	sweep_0/sweep_mode	Yes	string		['sector', 'coplane', 'rhi', 'vertical_pointing', 'idle', 'azimuth_surveillance', 'elevation_surveillance', 'sunscan', 'pointing', 'manual_ppi', 'manual_rhi', 'doppler_beam_swinging', 'complex_trajectory', 'electronic_steering']	azimuth_surveillance	Mandatory	pass
sweep_variables	sweep_0/follow_mode	Yes	string		['none', 'sun', 'vehicle', 'aircraft', 'target', 'manual']	not_set	Mandatory	fail_mandatory
sweep_variables	sweep_0/prt_mode	Yes	string		['fixed', 'staggered', 'dual', 'hybrid']	not_set	Mandatory	fail_mandatory
sweep_variables	sweep_0/fixed_angle	No	float	None	None	None	Mandatory	fail_mandatory
sweep_variables:sweep_0/fixed_angle	units	No	string	None	degrees	None	Mandatory	fail_mandatory
sweep_variables	sweep_0/azimuth	Yes	float	float64	None	47.5159912109375	Mandatory	fail_mandatory
sweep_variables:sweep_0/azimuth	units	Yes	string	str	degrees	degrees	Mandatory	pass
sweep_variables:sweep_0/azimuth	standard_name	Yes	string	str	sensor_to_target_azimuth_angle	ray_azimuth_angle	Mandatory	fail_mandatory
sweep_variables:sweep_0/azimuth	long_name	Yes	string	str	Azimuth angle from true north	azimuth_angle_from_true_north	Mandatory	fail_mandatory
sweep_variables:sweep_0/azimuth	axis	Yes	string	str	cradial_azimuth_coordinate	radial_azimuth_coordinate	Mandatory	pass
sweep_variables	sweep_0/elevation	Yes	float	float64	None	0.6	Mandatory	fail_mandatory
sweep_variables:sweep_0/elevation	units	Yes	string	str	degrees	degrees	Mandatory	pass
sweep_variables:sweep_0/elevation	standard_name	Yes	string	str	sensor_to_target_elevation_angle	ray_elevation_angle	Mandatory	fail_mandatory
sweep_variables:sweep_0/elevation	long_name	Yes	string	str	Elevation angle from horizontal plane	elevation_angle_from_horizontal_plane	Mandatory	fail_mandatory
sweep_variables:sweep_0/elevation	axis	Yes	string	str	cradial_elevation_coordinate	radial_elevation_coordinate	Mandatory	pass

Datasets

Group	Name	Available	Expected Dtype	Actual Dtype	Expected Value	Actual Value	Requirement	Result
data_variables	sweep_0/DBZH	Yes					dataset	pass
data_variables:sweep_0/DBZH	standard_name	Yes			radar_equivalent_reflectivity_factor_h	radar_equivalent_reflectivity_factor_h	Mandatory	pass
data_variables:sweep_0/DBZH	long_name	Yes			Equivalent reflectivity factor H	Equivalent reflectivity factor H	Mandatory	pass
data_variables:sweep_0/DBZH	wmo__parameter_url	No	string	None	None	None	Mandatory	fail_mandatory
data_variables:sweep_0/DBZH	wmo__parameter_name	No	string	None	None	None	Mandatory	fail_mandatory
data_variables:sweep_0/DBZH	valid_range	No		None	None	None	Mandatory	fail_mandatory
data_variables:sweep_0/DBZH	units	Yes			None	dBZ	Mandatory	pass
data_variables:sweep_0/DBZH	coordinates	Yes			None	elevation azimuth range latitude longitude altitude time	Optional	pass

Expert Team on Operational Weather Radar
Standing Committee on Measurements, Instrumentation and Traceability, WMO